Technical University of Denmark

SPRING, 2019
Campus: Lyngby

MODEL-BASED MACHINE LEARNING

Quiz 2

(Time allowed: 45 MINUTES)

Name: _____ Student ID: _____

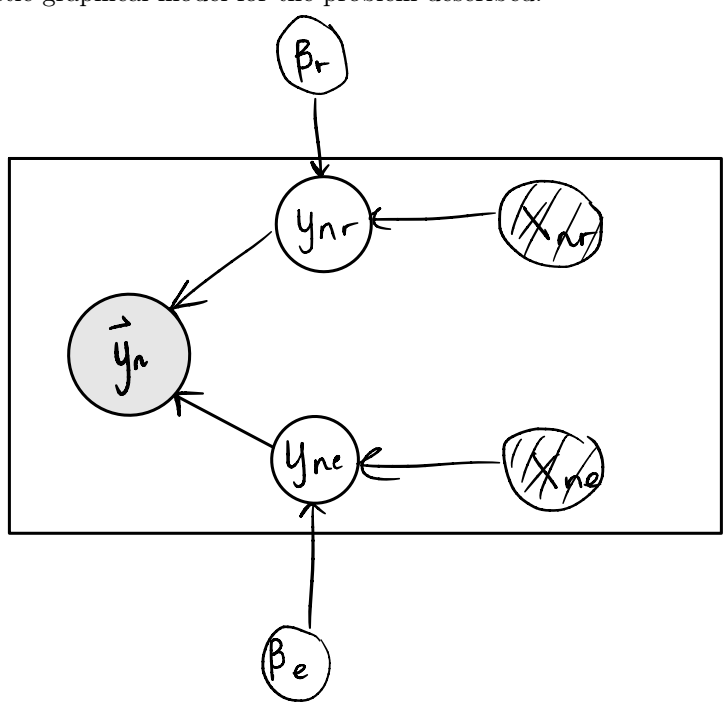
Part 1: Machine Learning Modeling

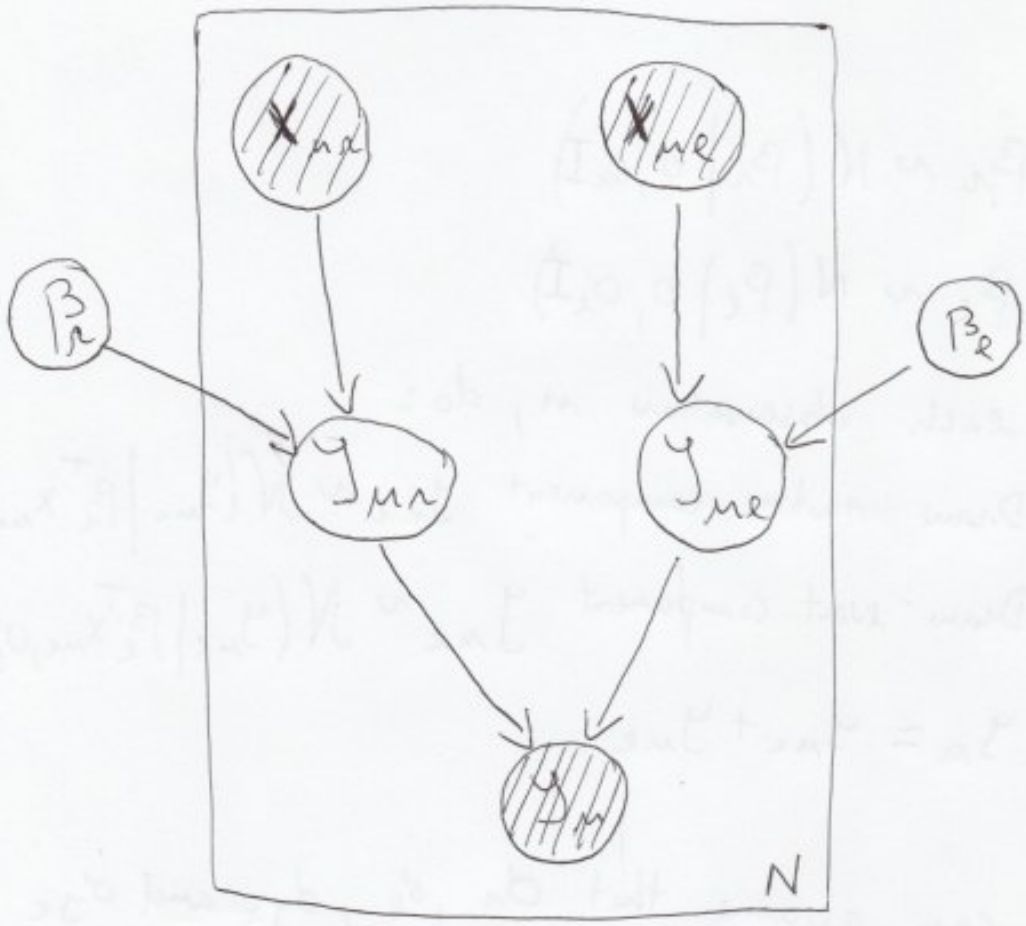
You have just received a dataset of 5 years of public transport travel demand to the Parken stadium area in Copenhagen **during event periods only**. It consists of values y_n, one per event, where each y_n corresponds to how many people went to that area by any public transport means during the event period.

Your domain knowledge tells you that, for each observation n, the total demand y_n corresponds to the number of people that went to the Parken area as part of their daily routine ("routine" component, y_{nr}, i.e. for people that live or work there, or go there normally) **plus** the number of people that went there for the event: y_{ne}. You assume that the routine component is a linear function of attributes $\mathbf{x}_{nr}$ that describe it (e.g. period the of day and day of week it represents), while the event component is a linear function of the event characteristics $\mathbf{x}_{ne}$: the category (e.g. sports, concert, others), facebook likes to that event, etc.

Your goal is to implement a probabilistic model that explains travel demand in the area, and allows you to infer the contribution of the events to the total observed demand y_n.

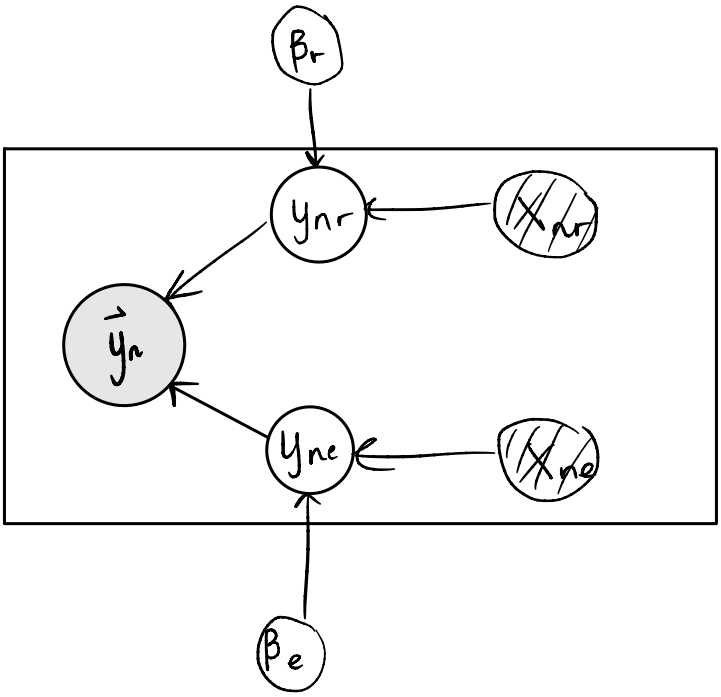
1. (20%) Draw the probabilistic graphical model for the problem described.





P.S. parameters of the model are ommited for clarity

2. (20%) Write a generative story consistent with your PGM. You may assume that all random variables are Normally distributed.



Draw parameter $\beta_r \sim \mathcal{N}(0, \sigma_r I)$

Draw parameter $\beta_e \sim \mathcal{N}(0, \sigma_e I)$

For each observation:

Draw routine component $\vec{y}_{nr} \sim \mathcal{N}(\vec{y}_{nr} \mid \beta_r^T \vec{X}_{nr}, \sigma_{nr})$

Draw event component $\vec{y}_{ne} \sim \mathcal{N}(\vec{y}_{ne} \mid \beta_e^T \vec{X}_{ne}, \sigma_{ne})$

Draw $\vec{y}_n \sim \mathcal{N}(\vec{y}_n \mid \vec{y}_{nr} + \vec{y}_{ne}, \sigma_n)$

Draw $\beta_r \sim N(\beta_r \mid 0, \sigma_r I)$

Draw $\beta_e \sim N(\beta_e \mid 0, \sigma_e I)$

For each observation n, do:

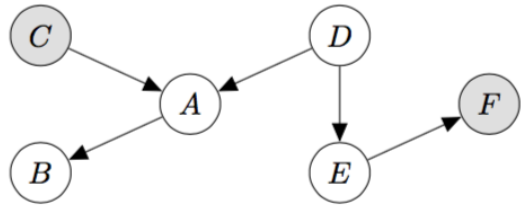
Draw routine component $y_{nr} \sim N(y_{nr} \mid \beta_r^T x_{nr}, \sigma_{yr}$

Draw event component $y_{ne} \sim N(y_{ne} \mid \beta_e^T x_{ne}, \sigma_{ye})$

$y_n = y_{nr} + y_{ne}$

P.S.: we can assume that σ_r, σ_e, σ_{yr} and σ_{ye} are given as parameters of the model.

Part 2: Inference - Belief Propagation



P(A|C,D)

	$A=0$	$A=1$
$C=0, D=0$	0.5	0.5
$C=0, D=1$	0.9	0.1
$C=1, D=0$	0.1	0.9
$C=1, D=1$	0	1

P(B|A)

	$B=0$	$B=1$
$A=0$	0.3	0.7
$A=1$	0	1

P(D)

$D=0$	$D=1$
0.8	0.2

P(E|D)

	$E=0$	$E=1$
$D=0$	0.7	0.3
$D=1$	0.8	0.2

P(C)

$C=0$	$C=1$
0.7	0.3

P(F|E)

	$F=0$	$F=1$
$E=0$	0.2	0.8
$E=1$	0.7	0.3

Figure 1: A Probabilistic Graphical Model and its Conditional Probability Tables

Consider the PGM in Figure 1, together with some quantities already calculated (in Figure 2 below).

(a) (60%) Calculate $Bel(E)$

Please include all intermediate steps (i.e. all π and λ messages and beliefs, renormalization).

$\pi(A)$

$A=0$	$A=1$
0.58	0.42

$\pi(B)$

$B=0$	$B=1$
0.174	0.826

$\lambda(A) = \lambda(B) = 1$

$\pi(C) = \lambda(C)$

$C=0$	$C=1$
1	0

$\pi(D) = P(D)$

$D=0$	$D=1$
0.8	0.2

$\pi(F) = \lambda(F)$

$F=0$	$F=1$
1	0

Figure 2: Pre-calculated beliefs

$$Bel(E) \propto \pi(E)\,\lambda(E)$$

To determine $\pi(E)$ and $\lambda(E)$, we need to determine the messages $\pi_{D,E}(D)$ and $\lambda_{F,E}(E)$.

$$\lambda_{F,E}(E) = \sum_F \lambda(F)\, P(F|E)$$

We know that $\lambda(F=0)=1$ and $\lambda(F=1)=0$, from tables in Figure 2.

$$\lambda_{FE}(E) = P(F=0|E) \Longrightarrow$$

$\lambda_{FE}(E)$

$E=0$	$E=1$
0.2	0.7

We can already calculate $\lambda(E)$:

$$\lambda(E) = \lambda_{FE}(E) \quad \text{(E has only one child)}$$

To determine $\pi_{DE}(D)$, we need to propagate the messages to D. Some direct calculations, by checking the tables in Figure 2 (and the $P(A)$):

$$\lambda_{BA}(A) = \lambda(B) = 1$$

$$\pi_{CA}(C) = \pi(C)$$

$$\lambda_{AD}(D) = \sum_A \lambda(A) \sum_C P(A|C,D)\, \pi_{C,A}(C) =$$

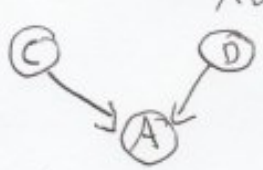
Notice that $\lambda(A)=1$, from the tables

$$= \sum_{A,C} P(A|C,D)\,\pi(C) =$$

we know that $\pi(C=0)=1$, $\pi(C=1)=0$

$$= \sum_A P(A|C=0,D) = \underbrace{P(A=0|C=0,D) + P(A=1|C=0,D)}_{=1}, \text{ i.e. } \lambda_{AD}(D)=1$$

Note: we could know that $\lambda_{AD}(D) = 1$ (i.e $D \perp C$) just by seeing that



is a V-structure!

In practice we would not NEED to CALCULATE $\lambda_{AD}(D)$ for that REASON, UNLESS A WAS OBSERVED.

$$\pi_{DE}(D) = \pi(D)\,\lambda_{AD}(D) = \pi(D)$$

In table in Figure 2, $\pi(D) = P(D)$

$$\pi(E) = \sum_D P(E|D)\,\pi_{DE}(D) = \sum_D P(E|D)\,P(D) =$$

$$= P(E|D=0)P(D=0) + P(E|D=1)P(D=1)$$

$E=0$

$$\pi(E=0) = P(E=0|D=0)P(D=0) + P(E=0|D=1)P(D=1) = 0.7 \times 0.8 + 0.8 \times 0.2 = 0.72$$

$E=1$

$$\pi(E=1) = 1 - \pi(E=0) = 0.28$$

$\pi(E)$

$E=0$	$E=1$
0.72	0.28

$$Bel(E) \propto \pi(E)\,\lambda(E)$$

$E=0$ $\quad \pi(E=0)\,\lambda(E=0) = 0.72 \times 0.2 = 0.144$

$E=1$ $\quad \pi(E=1)\,\lambda(E=1) = 0.28 \times 0.7 = 0.196$

Normalization constant $= \dfrac{1}{0.144 + 0.196} = 2.94$

$Bel(E)$

$E=0$	$E=1$
0.42	0.58